Java String Lab Solutions (Level 1, 2 & 3)

Level 1 - Compare Two Strings

```
import java.util.*;
class CompareStrings {
    public static boolean compare(String a, String b) {
        if (a.length() != b.length()) return false;
        for (int i = 0; i < a.length(); i++) {
            if (a.charAt(i) != b.charAt(i)) return false;
        }
        return true;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s1 = sc.next();
        String s2 = sc.next();
        System.out.println(compare(s1, s2));
        System.out.println(s1.equals(s2));
    }
}
```

Level 1 - Substring using charAt

```
import java.util.*;
class SubstringDemo {
    public static String getSub(String s, int start, int end) {
        String res = "";
        for (int i = start; i < end; i++) {
            res += s.charAt(i);
        }
        return res;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.next();
        int start = sc.nextInt();
        int end = sc.nextInt();
        System.out.println(getSub(s, start, end));
        System.out.println(s.substring(start, end));
    }
}
```

Level 1 - Convert to Uppercase

```
import java.util.*;
class UpperCaseDemo {
    public static String toUpper(String s) {
        String res = "";
        for (int i = 0; i < s.length(); i++) {
            char ch = s.charAt(i);
            if (ch >= 'a' && ch <= 'z')
                ch = (char)(ch - 32);
            res += ch;
        }
        return res;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        System.out.println(toUpper(s));
        System.out.println(s.toUpperCase());
    }
}
```

Level 2 - Find Length Without length()

```
import java.util.*;
class StringLength {
    public static int findLength(String str) {
        int count = 0;
        try {
            while (true) {
                str.charAt(count);
                count++;
            }
        } catch (Exception e) {
            return count;
        }
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String text = sc.nextLine();
        System.out.println(findLength(text));
        System.out.println(text.length());
    }
}
```

Level 2 - Split Text into Words

```
import java.util.*;
class SplitWords {
    public static String[] splitText(String text) {
        ArrayList<String> words = new ArrayList<>();
        String word = "";
        for (int i = 0; i < text.length(); i++) {
            if (text.charAt(i) != ' ')
                word += text.charAt(i);
            else {
                words.add(word);
                word = "";
            }
        }
        words.add(word);
        return words.toArray(new String[0]);
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String text = sc.nextLine();
        String[] result = splitText(text);
        for (String w : result)
            System.out.println(w);
    }
}
```

Level 2 - Count Vowels & Consonants

```
import java.util.*;
class VowelConsonant {
    public static int[] countVC(String text) {
        int v=0,c=0;
        for(int i=0;i<text.length();i++){
            char ch=Character.toLowerCase(text.charAt(i));
            if(Character.isLetter(ch)){
                if("aeiou".indexOf(ch)!=-1) v++;
                else c++;
            }
        }
        return new int[]{v,c};
```

```
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String text=sc.nextLine();
        int[] res=countVC(text);
        System.out.println("Vowels: "+res[0]);
        System.out.println("Consonants: "+res[1]);
    }
}
```

Level 3 - First Non-Repeating Character

```
import java.util.*;
class FirstNonRepeat {
    public static char findChar(String text){
        int[] freq=new int[256];
        for(int i=0;i<text.length();i++)
            freq[text.charAt(i)]++;
        for(int i=0;i<text.length();i++)
            if(freq[text.charAt(i)]==1)
                return text.charAt(i);
        return '\0';
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String text=sc.nextLine();
        char res=findChar(text);
        if(res!='\0') System.out.println(res);
        else System.out.println("None");
    }
}
```

Level 3 - Palindrome Check

```
import java.util.*;
class PalindromeCheck {
    public static boolean isPalindrome(String text){
        int s=0,e=text.length()-1;
        while(s<e){
            if(text.charAt(s)!=text.charAt(e))
                return false;
            s++; e--;
        }
        return true;
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String text=sc.nextLine();
        if(isPalindrome(text))
            System.out.println("Palindrome");
        else
            System.out.println("Not Palindrome");
    }
}
```